

NILOY SAHA

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SUMMARY

Computer Science student with experience building scalable infrastructure automation, observability systems, and real-time AI applications. Skilled in Python, backend development, distributed monitoring, and low-latency system design, with hands-on experience improving operational efficiency across 5,000+ servers.

EDUCATION

Hofstra University

New York, USA

B.S. in Computer Science

Expected Graduation, Dec 2027

- o **GPA:** 3.46/4.00, *Dean's List*
- o **Related Coursework:** Data Structures & Algorithms, Computer Architecture & Programming, Software Engineering, Object-Oriented Programming, Applied NLP and Text Mining, Operating Systems, Building RAG Agents with LLM (Course By NVIDIA), Machine Learning, Intro to Artificial Intelligence

EXPERIENCE

Optimum Communications INC.

New York, USA

Software Developer Intern

Jan 2026 - Current

- **Automated infrastructure observability** across **5,000+ servers** by configuring Zabbix agents and SNMP polling to collect **10+** real-time system and container metrics per host.
- **Engineered dynamic Low-Level Discovery (LLD) pipelines** to auto-monitor **8+ CPU cores per machine** across **5,000+ servers**, reducing manual monitoring configuration effort by **70%**.
- **Developed** a custom **Bash-based Docker discovery framework** that auto-detected and monitored containers across **5,000+ servers**, achieving **100% automation of container observability workflows**.
- **Designed trigger-based alerting and incident response pipelines** across **5,000+ monitored servers**, reducing simulated mean-time-to-detection (MTTD) by over **50%**.

Maurice A. Deane School of Law at Hofstra University

New York, USA

IT Technician

Sept 2025 - Dec 2025

- **Automated log analysis and diagnostics** in Python with Pandas, cutting resolution time by **15%**.
- **Maintained 99% uptime** across **50+ endpoints** through proactive monitoring and scripting.
- **Developed** a searchable **SQL-based knowledge base**, reducing repeat technical issues by **20%**.
- **Implemented** automated **TCP/IP and DNS testing scripts**, reducing manual network checks by **30%** and improving connectivity reliability.

PROJECTS

My Jarvis (Autonomous Local AI Agent) | GitHub: <https://github.com/NiloySaha84/myJarvis>

- **Built** a local **AI desktop assistant** using LangGraph, FastAPI, Electron, and Deepgram to autonomously execute multi-step workflows, achieving **~4.8s average LLM latency** and **~10.5s end-to-end voice response time**.
- **Engineered** a **real-time streaming STT/TTS pipeline** that processed **22K+ audio chunks per session** and **eliminated recursive voice feedback** through transcript filtering, microphone muting, and concurrency-safe turn synchronization.
- **Developed** a **hybrid RAG retrieval system** with ChromaDB, BM25, and FlashRank reranking to enable **workflow-aware automation from local knowledge**, reducing retrieval latency to **~1.7s per query**.
- **Implemented runtime observability** across agent, API, retrieval, and voice subsystems, tracking **40+ operational metrics** including tool execution success, WebSocket latency, Deepgram stream health, and indexing/query performance for faster debugging and reliability analysis..

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C++

Frameworks & Libraries: NextJs, NodeJs, Matplotlib, Pandas, NumPy, Scikit-Learn, Huggingface, LangChain, LangGraph

Databases: PostgreSQL, MongoDB, Supabase, VectorDBs (e.g. FAISS)

Cloud & DevOps: Docker, Vercel, CI/CD Pipelines

Core Competencies: RESTful APIs, Data Structures & Algorithms, OOP, NLP, ML/AI, RAG Architecture, Agentic AI, MCP

AI Tools: Claude Code, Cursor, Codex